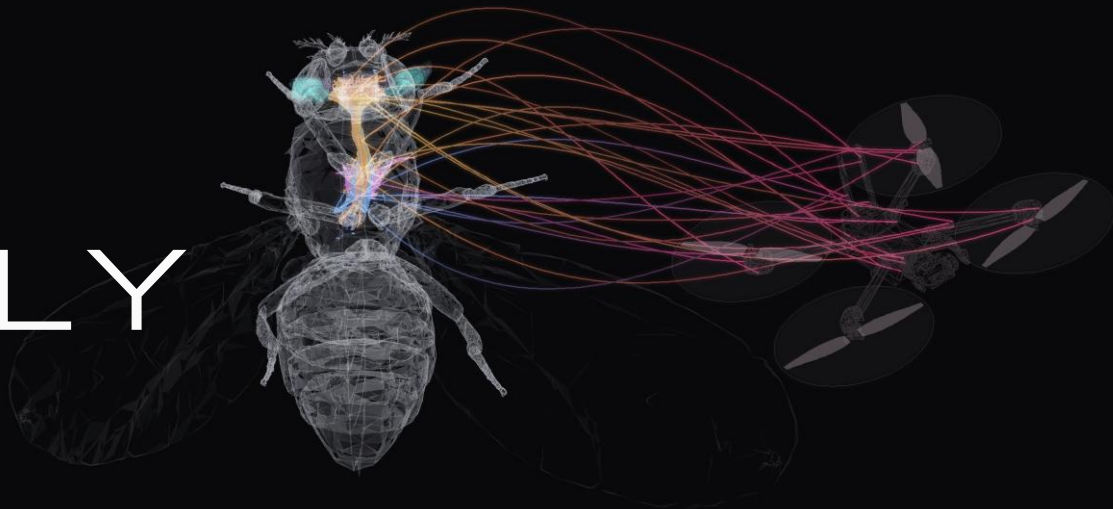




BANC V888 CONNECTOME

NEUROFLY

The brain of a fruit fly which
pilots a drone



FULL BANC V888 CONNECTOME

175,401 NEURONS

CLOSED LOOP IN MUJOCO

WEIGHTS MEASURED, NOT TRAINED

TYPICAL AI

Random weights, learned

Millions of examples to train, and nobody can say what one weight means.



Connectome: the complete wiring diagram of a nervous system

NEUROFLY

Real fly wiring, measured

Every synapse mapped in a real fruit fly. Evolution tuned it for 400 million years, and each neuron has a name.

18.6 M

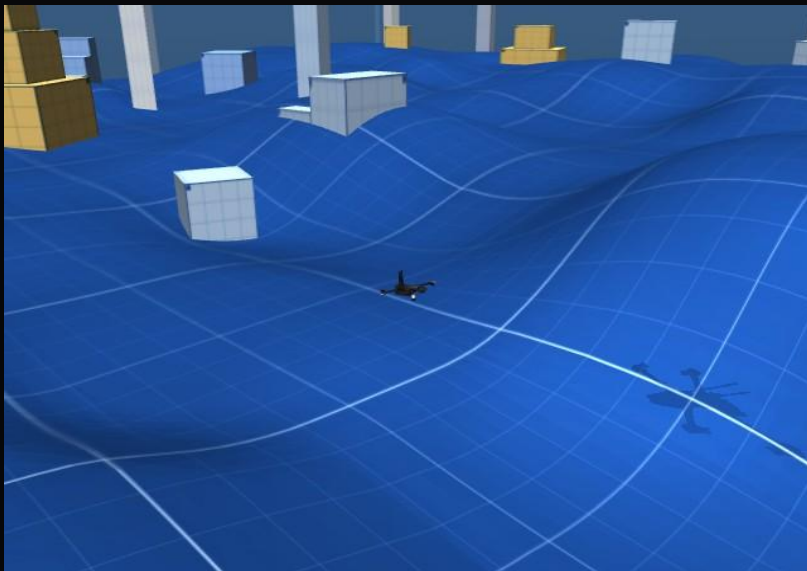
real synapses in the controller

0

connections or weights we invented

BANC v888: first connectome of a whole fly brain plus nerve cord (Bates et al. 2026)

USE CASE: THE LAST METRES



Simulated world: hills, blocks, wind, a target mast

A rescue drone homes in on a beacon among rubble or trees, where GPS drifts by metres. GPS brings it close. The last 4 m are steered by the fly brain, from a plain camera.

far: GPS

near: fly brain

2.5 m

GPS error in our simulation

100%

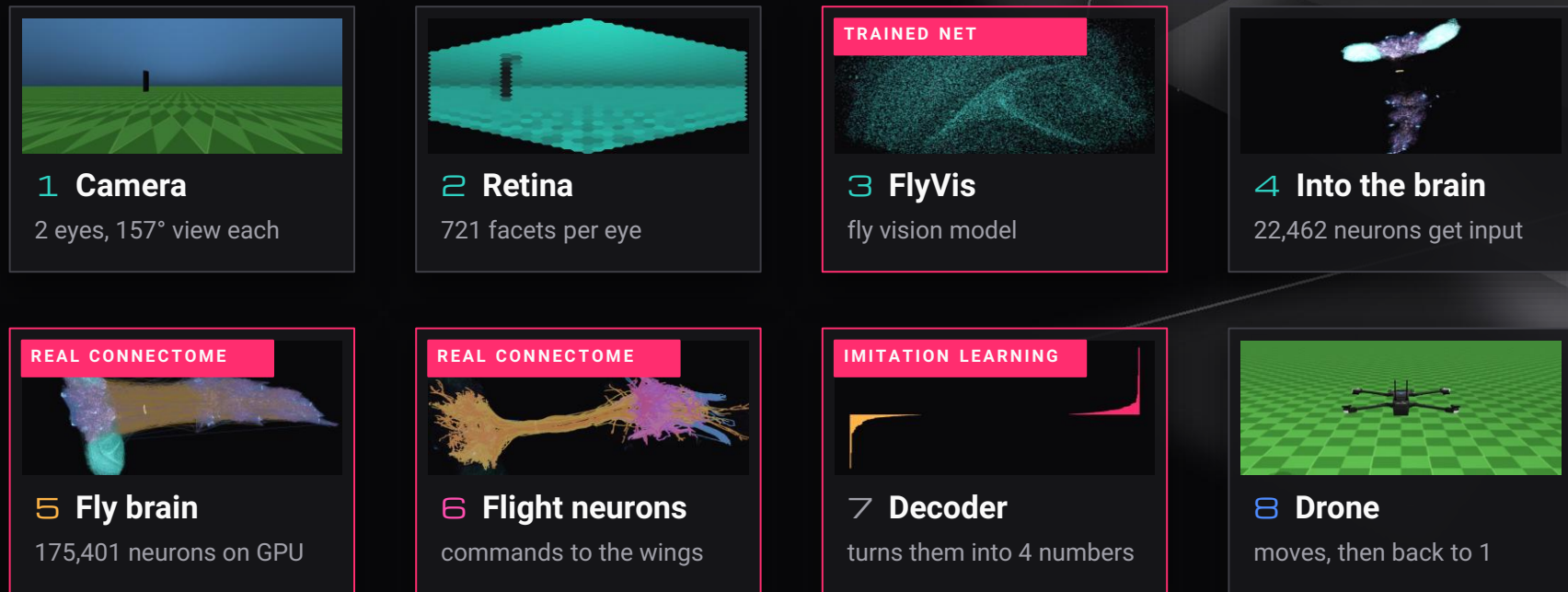
of the final heading comes from fly neurons

Beacon: the target the drone must reach, here a dark mast

Heading: the direction the nose points

Status: shown in simulation, not yet on a real drone

HOW IT WORKS: ONE LOOP, 8 STEPS

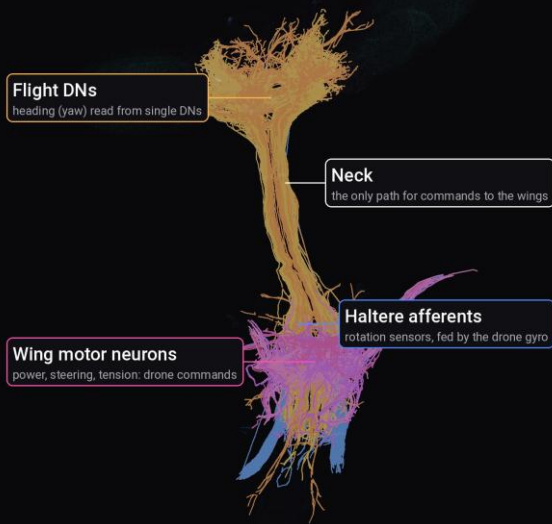


Red tag: where AI works in the loop

FlyVis: published model of the fly eye and visual brain

Speed: 2.6 ms per frame for the whole connectome on a laptop GPU

THE BRAIN: REAL, NOT INVENTED



175,401

neurons in the fly brain and nerve cord

1.53 M

neuron-to-neuron connections we use

375

flight neurons from the brain

60

wing muscle neurons

428

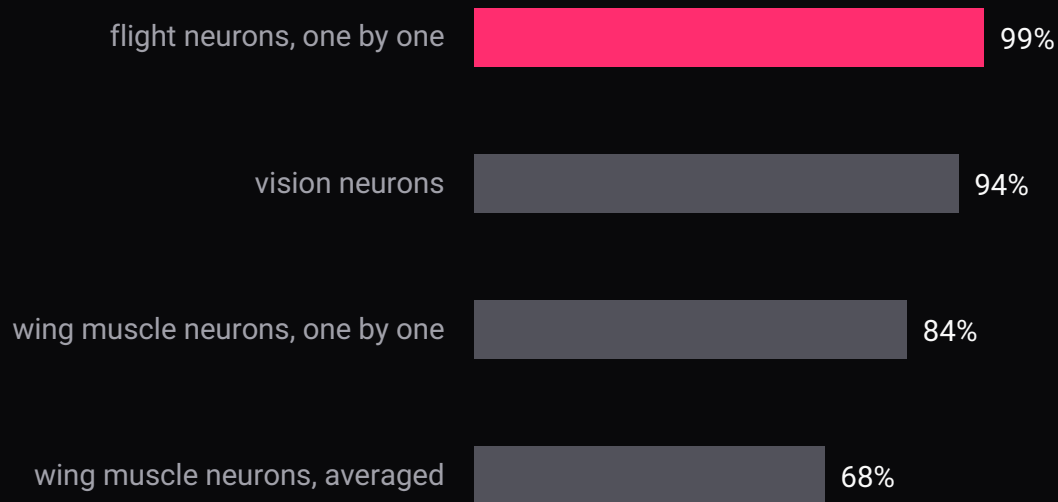
haltere neurons, the fly's gyroscope

Every group comes from official BANC labels. We never guess a neuron's name or add a connection.

Render: 863 flight neurons in full 3D shape, from BANC skeletons

Haltere: tiny hind wing that senses rotation

SINGLE NEURONS KNOW THE TARGET



99%

of scenes where single flight neurons
tell left from right correctly

68%

if you average the muscle neurons: left
and right cancel out

So we read the brain neuron by neuron.

Accuracy: how often we can tell which side the target is on (50% = guessing)

RESULTS: IT LEARNS TO FLY TO THE TARGET

25.1° to 2.65°

average heading miss, before vs
after 1,000 training flights

138 / 150

flights reached the target with no
teacher at all (92%)

18.2 m to 2.1 m

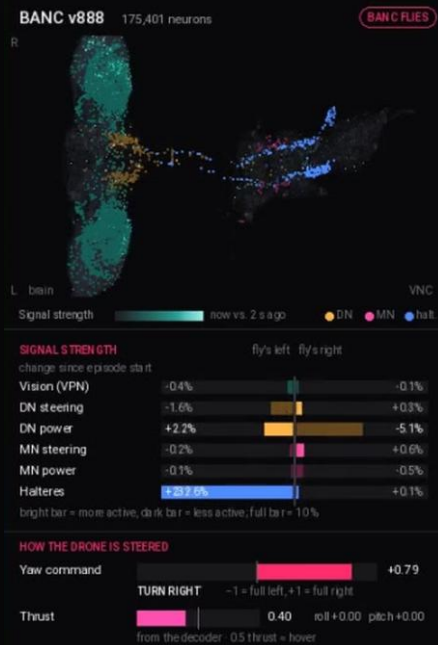
closest approach in 6 new worlds
never seen in training

35° to 1.5°

pushed off course by a gust, back on
course in 3.9 s**Heading miss:** angle between the nose and the target, 0° = straight at it**Bottom right:** what the drone's two eyes see, the target is the dark mast

RESULTS

YOU STAY IN CONTROL



1 See who steers

Live share of the heading from the fly brain vs GPS, plus the final command.

2 See why it turns

Each flight neuron's vote for left or right, from the real decoder weights.

3 Take over any time

Sensors keep the drone level, GPS takes over far away, a pilot can fly by keyboard while the brain only watches.

4 Check every neuron

3D explorer: every neuron, its class and its real synaptic partners from BANC.

Left: live BANC panel in flight, a brain you can watch

Not a black box: every command traces back from pixel to neuron to wing

HONEST LIMITS AND NEXT STEPS

WHAT DOES NOT WORK YET

Lift	fly-brain thrust too weak, the simulator holds height
Left eye	fewer labelled neurons in BANC, $\sim 3\times$ less input
Obstacles	the fly brain does not avoid blocks yet
Hardware	simulation only, no real drone flown

From BANC: neurons, synapses, excite or inhibit, neuron groups, 3D shapes

Our assumptions: the neuron activity equation, how vision enters, how flight neurons become drone commands

TODAY

demonstrated

Full BANC steers a simulated drone in random worlds

NEXT

planned

Two cameras on a micro-drone, connectome on an onboard GPU

LATER

potential

Search and rescue homing, low-power neuromorphic chips

EVOLUTION WROTE THE WEIGHTS. WE GAVE THEM WINGS.

TEAM THE ROOK

The eyes camera, fly retina, FlyVis, input to BANC

The brain full connectome on GPU, decoders, training

Body and world drone, sensors, wind, random worlds

Dawid Wypych Mateusz Nowaczek Krzysztof Mazur

BUILT ON, WITH THANKS

BANC v888 connectome
Bates et al. 2026, doi 10.7910/DVN/7WTH1N

FlyVis, FlyGym, MuJoCo, Skydio X2 model
open models and simulator (MuJoCo Menagerie)

Claude Code
AI assistant for code, docs and slides

Built during HackYeah: the whole loop, controller, decoders and training, drone simulator, live panel, 3D explorer.

[GITHUB.COM/FALCONDEVX/NEUROFLY](https://github.com/falcondevx/neurofly)